
Ergonomics

1.0 BACKGROUND

Ergonomic issues are a growing area of concern to all segments of the industrial world. There is an exponentially increasing awareness of the contribution that repetitive manual tasks have to the incidence of upper extremity and low back problems. The semiconductor industry is not immune to these same ergonomic hazards. The U.S. Bureau of Labor Statistics has identified that, in 1990, 1991 and 1992, occupational illness incidence rates for disorders associated with repeated trauma for semiconductors and related device manufacturing (SIC 3674) were:

Year	Incidence Rate	Percent Relative Standard Error
1990	24.9 per 10,000 full-time workers (FTW)	29
1991	45.8 per 10,000 FTW	11
1992	47.7 per 10,000 FTW	13

Figure 7.1 identifies the latest data available from the Semiconductor Industry Association's OHS Database for 1982–1992 on the comparative incidence of strains/sprains vs. *cumulative trauma disorders* (CTD).^[2]

Ergonomics, also known as human factors engineering, is a multi-functional discipline concerned with the design of equipment, operations, and work environments to match human capabilities and limitations.

Virtually every aspect of equipment design impacts the user interface in one manner or another. The acceptability of a semiconductor fab area or process tool may depend heavily on how well the fab area operators are able to perform their assigned task utilizing the equipment, tools, and work areas provided. The overall intent of an ergonomic evaluation is to focus upon and resolve human-equipment interface problems, finding solutions which promote system effectiveness.^{[3][4]}

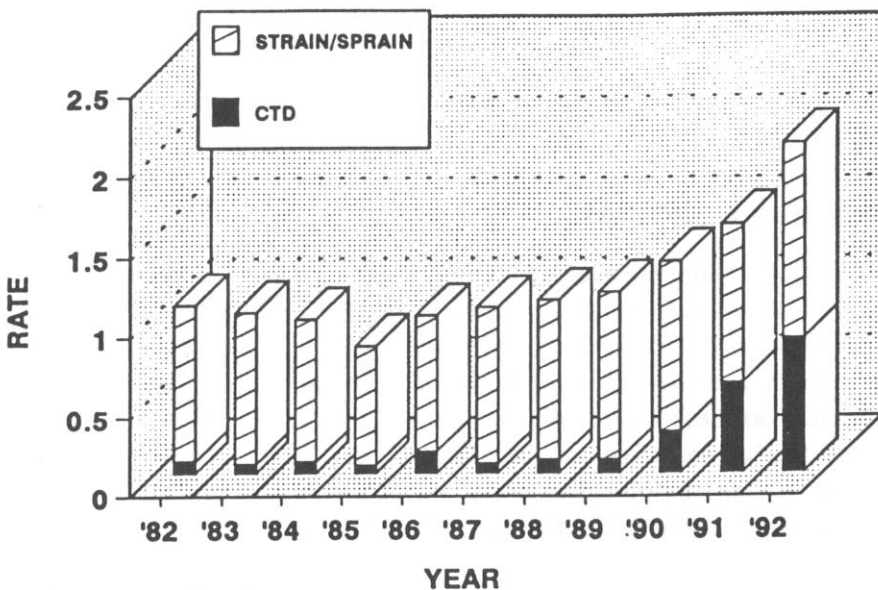


Figure 7.1. The Semiconductor Industry Association's OHS Database for 1982–1992: the comparative incidence of strains/sprains vs. cumulative trauma disorders (CTD).^[2]

2.0 IDENTIFICATION AND EVALUATION OF PHYSICAL STRESSORS

The identification and evaluation of the physical stressors to which a worker may be exposed are extremely important in attempting to understand the “source” of cumulative trauma disorders, and more specifically low back pain. The cumulative trauma disorder (CTD) and low back pain risk factors are:

- Posture
- Forceful exertions
- Repetition
- Vibration
- Temperature

Each of these risk factors warrant further discussion.

2.1 Posture

The posture required when performing a task can significantly increase the risk of CTD or low back problems. The most important dynamic postures are bending forward, twisting, and reaching. Bending, particularly deep bending in a stooped posture, exerts stress on the back from supporting the weight of the upper body and from stabilizing the body. When combined with lifting an object, this posture can be very hazardous. Low bending lifts have been implicated in the majority of low back injuries with medical treatment. Twisting applies additional stress on the lower back. Twisting when lifting has been implicated in approximately 20% of low back cases requiring medical treatment. Twisting occurs when the shoulders are moved out of normal alignment with the hips. Reaching out to lift an object stresses the lower back in proportion to the distance of the reach. By holding an object away from the body, the risk of injury increases substantially. This is because the lower back must use greater muscle contractions to support the load. These contractions also compress the inter-vertebral discs. Static postures may also lead to CTD and low back pain. Seated or standing static postures do not put muscles completely at rest. Muscles that serve to stabilize the body are held in tension and begin to fatigue fairly quickly. Seated, unsupported or minimally supported seated postures result in a near doubling of disc pressure compared to a standing posture.^[5] Static postures reduce blood flow to the muscles.

2.2 Forceful Exertions

The exertions that require significant force include lifting, lowering, pushing, pulling, and carrying that put stress on the back or limbs. The more force that is required, the more stress is sustained by the body. Lifting is especially stressful when it begins below knee level, or ends or starts above shoulder level. Catching dropped, thrown, or falling objects also involves

forceful exertions. Back muscles tense appreciably in preparation for absorbing the load. The degree of tension is a function of preview time, knowledge or lack of knowledge of the weight of the object, and velocity of the object.^[5]

2.3 Repetition

The actual frequency with which manual materials handling or job tasks are accomplished over a defined time period is known as the rate of repetition. The higher the repetition rate, the more demand is placed on the cardiovascular system. Low frequency tasks provide sufficient time for the body or limb to recover. Tasks of high repetition often do not allow sufficient recovery time and fatigue or other micro-injuries develop.^[5]

2.4 Mechanical Vibration

Vibration is the transfer of mechanical energy passing through the body as oscillations. If the vibration is acting on the body axially, from the floor or through a seat for example, the oscillations affect the discs in the spinal column. The effect is dependent on the amplitude and frequency of the vibration. Most important is vibration in the 10–12 Hz frequency range as there is some resonance effect in the spine.^[5]

2.5 Temperatures

High ambient temperatures raise the internal body temperature. Fatigue occurs more quickly and there can also be dehydration. Conversely, low ambient temperatures restrict blood flow to the core of the body, and can contribute to injuries due to the lack of blood flow to the tissues.^[5]

This brief discussion of the four primary stressors in most cumulative trauma disorder cases is intended to be general in nature. If we look at a more defined medical condition such as wrist injury or carpal tunnel syndrome, some possible recommendations for modification of the “source” can be made. Table 7.1 identifies ergonomic intervention strategies for wrist injury (e.g., carpal tunnel syndrome) prevention.^[6] Numerous possible recommendations for dealing with the four primary physical stressor discussed are presented.

Table 7.1. Ergonomic intervention strategies for wrist injury prevention.^[6]
(Hodson, W., Maynard’s Industrial Engineering Handbook, © 1992, McGraw-Hill, reprinted with permission.)

Problem	Possible recommendations
1. Repetitiveness	1. Use mechanical aids 2. Enlarge work content 3. Rotate workers 4. Spread work uniformly across workshift 5. Automation
2. Large joint forces	1. Increase mechanical advantage of tools 2. Decrease the weight of tools, containers and parts 3. Increase the friction between handles and the hand 4. Optimize size and shape of handles 5. Select gloves to minimize effects on performance 6. Balance hand-held tools and containers 7. Use torque control devices 8. Fine tune work/rest ratios 9. Mechanical assists 10. Automation
3. Large, hand/arm contact forces	1. Enlarge corners and edges 2. Pads and cushions 3. Tool size selection
4. Poor posture	1. Locate work properly 2. Orient work properly 3. Select tool design for workstation 4. Change work method 5. Postural biofeedback training
5. Vibration	1. Select tools with minimum vibration 2. Select process to minimize surface and edge finishing 3. Mechanical assists 4. Use isolation for tools that operate above resonance point 5. Provide damping for tools that operate at resonance point 6. Vibration damping gloves 7. Adjust tool speed to avoid resonance

3.0 STRESSFUL POSTURES ASSOCIATED WITH CTD'S^{[7][8]}

Stressful postures constitute one of the primary occupational risk factors for the development of CTD's of the upper extremities. Postures are considered stressful when the body's position requires the use of force or causes stretching or compression of body tissues. The various work tasks need to be analyzed to see whether they require prolonged or repeated exertions involving the shoulder, wrist, and hand positions.

3.1 Shoulder

Jobs that require the worker to habitually reach and work the arms above the shoulder level or with elbows away from the body can influence the development of shoulder problems. Shoulder alignment can also be caused by repeatedly reaching behind the body or throwing materials over the shoulder.

3.2 Elbow

Rotation of the forearm, especially with the wrist bent, is particularly stressful to the elbow. Inward rotation, which turns the palm downward is called pronation. Outward rotation, which turns the palm upward, is called supination.

3.3 Wrist

Stressful wrist postures include extreme flexion or extension, and radial or ulnar deviation. These movements are particularly likely to cause CTDs when combined with forceful exertion such as a tight grip on a tool or other object.

3.4 Hand

There are three principle grips: power, hook, and pinch (or precision) grip. Avoid the repeated use of a pinch grip: it takes more strength and puts more tension on the tendons to hold an object between the fingers and thumb than it does to use a power or hook grip. The power grip is preferred, though it may be harmful if combined with frequent motion and

excessive force. The hook grip (such as holding a tray with your fingertips) is less stressful than the pinch grip, but not as good as the power grip.

3.5 Carpal Tunnel Syndrome^[9]

Generally, working repetitively in stressful postures results in added stress on the musculoskeletal system, and then the nervous system. In the case of carpal tunnel syndrome, this results in damage to the median nerve, leading to numbness, pain, and decreased motor function of the hand. The median nerve provides sensory distribution to the palmar surface of the hand, including the thumb, index, middle, and half of the ring finger. It also supplies motor function to the lumbrical muscles in the fingers. Figures 7.2 and 7.3 identify the sites of nerve entrapment causing numbness in the thumb and index finger (Fig. 7.2), and ring and little fingers (Fig. 7.3).^[10]

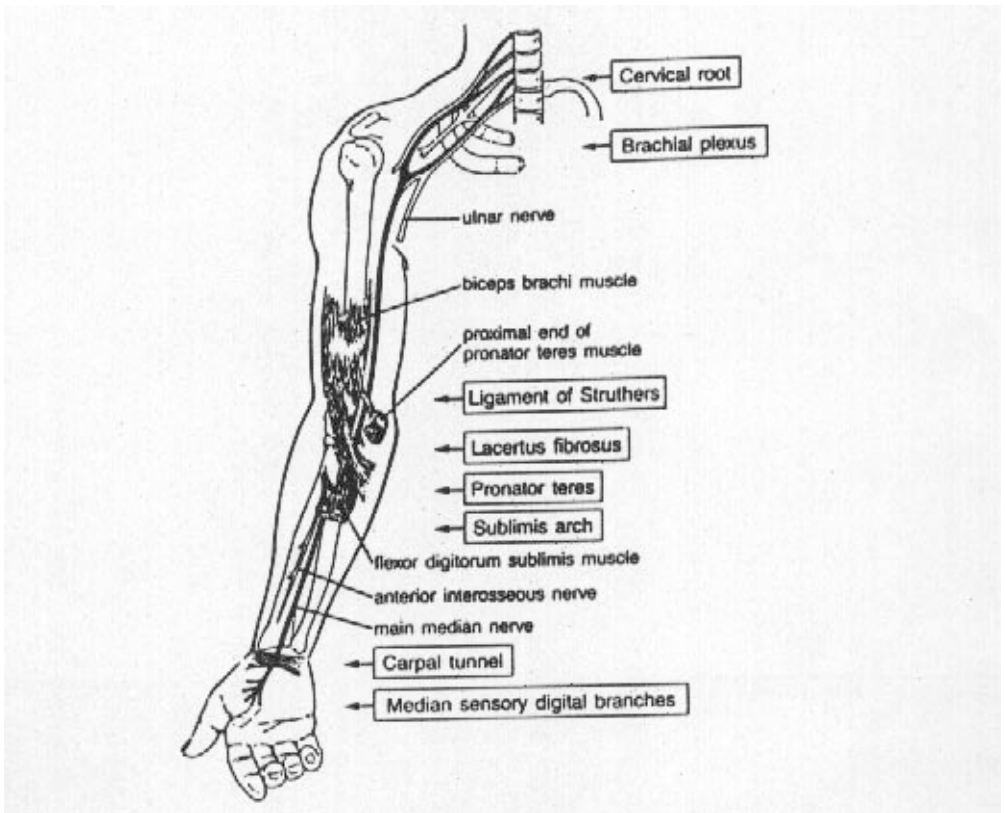


Figure 7.2. Sites of nerve entrapment causing numbness in the thumb and index fingers.^[9]
(© 1986, Hanley & Belfus, reprinted with permission.)

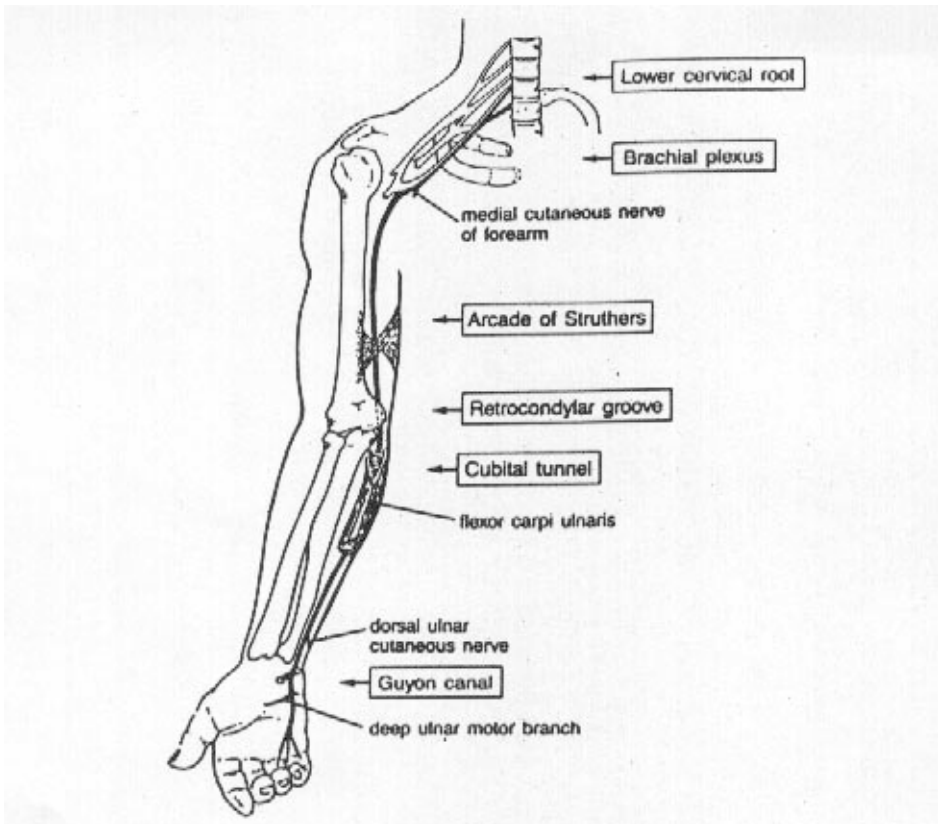


Figure 7.3. Sites of nerve entrapment causing numbness in the ring and little fingers.^[9]
(© 1986, Hanley & Belfus, reprinted with permission.)

Workers with carpal tunnel syndrome initially experience vague tingling or numb sensations in varying places on the front of the hand. Often they will be awakened at night with nocturnal numbness, a sensation so severe that it rouses them from sleep. As the problem progresses, the numbness and tingling may become fixed on certain locations on the hand, particularly the tips of the thumb, index, and middle fingers. Later, pain may develop, and the person may experience difficulty grasping objects. Finally, the motor fibers are affected, and the muscles themselves will become weak and will atrophy as the nerve damage continues.

One case-controlled study looking at cases of carpal tunnel syndrome in eleven industries found a statistically significant association with electronics assembly work, rapid repetitive hand movements, particularly in

the pinch and grasp positions, and the use of hand tools, particularly screwdrivers. One early concern of a similar nature within the semiconductor industry was the widespread use of special tweezers to hold and move wafers. Because the tweezers tended to damage the edge of the wafer as well as because of ergonomic concerns, the tweezers were generally replaced with the vacuum wand in the late 1970s to early 1980s.

The pathiophysiology of work-related carpal tunnel syndrome is thought to be based on several mechanisms, and workers may be at risk from one or more simultaneously. The carpal tunnel in the wrist is a narrow space surrounded by fixed unyielding structures, the carpal bones on the posterior surface of the wrist, and the transverse carpal ligament crossing the wrist on the anterior surface. Through this tunnel pass the flexor tendons of the hand, and the median nerve, with the later being the most compressible and vulnerable structure in the tunnel.

Rapid repetitive motion, particularly involving flexion of the wrist in the pinch and grasp position, may directly compress the median nerve against the transverse carpal ligament and may cause low-grade inflammation of the structures, increasing the pressure in the tunnel, and resulting in median nerve compression. Figure 7.4 provides a cross-sectional view of the wrist and the pertinent structures.^[11]

3.6 Other Potentially Contributing Factors

One study has noted that employees need to be aware of contributory factors other than work to ergonomic injuries.^[12] Their research and other studies have shown:

- Women are twice as likely to become a victim of a CTD than men. This is probably the result of women, in general, having smaller joint structure in comparison with men, and therefore more prone to these types of injuries.
- Other contributing factors can be diabetes, a lack of vitamin supplements (such as calcium, iron, and B-6), and menopause, all of which can increase an employees risk of carpal tunnel syndrome (CTS).
- Pregnant women may be more susceptible to CTD disorders because of fluid retention during pregnancy.

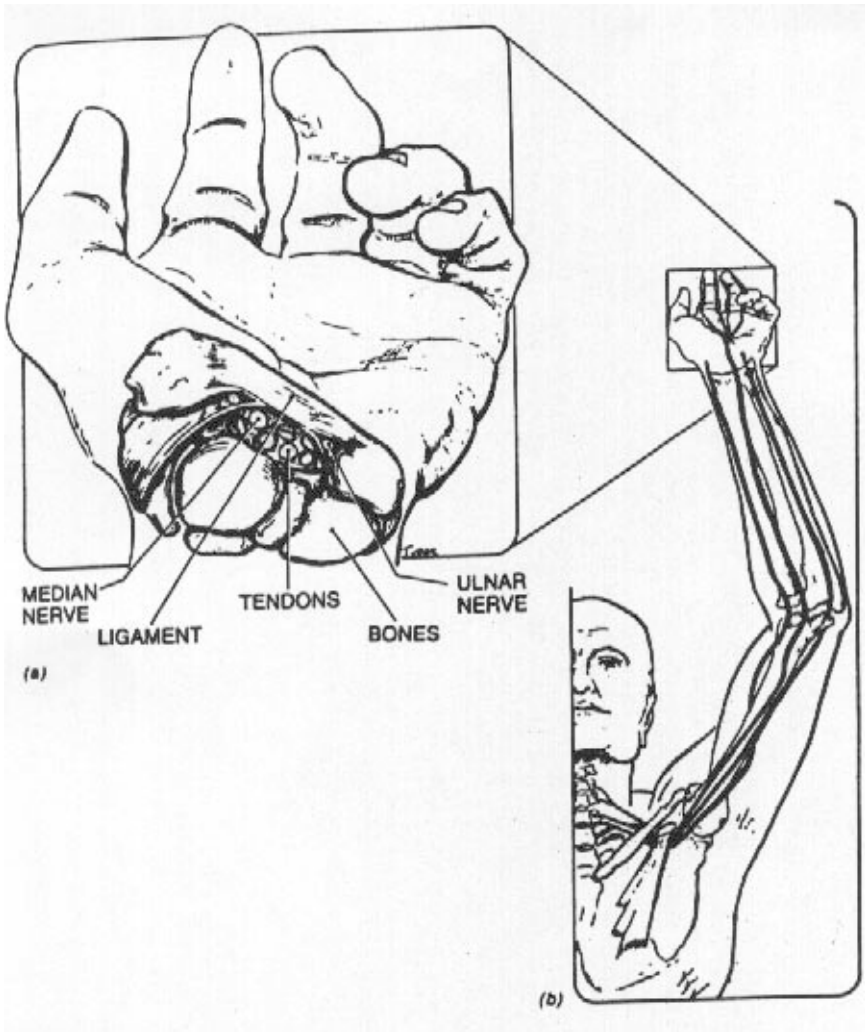


Figure 7.4. Cross-sectional view of the wrist and underlying structures.^[11] (© 1988, Taylor & Francis, reprinted with permission.)

4.0 SEMICONDUCTOR ERGONOMIC/MATERIAL HANDLING EXPOSURES

The primary areas where repetitive motion and manual materials-handling injuries occur are in the fab area, line maintenance support area, shipping/receiving docks, facilities maintenance shops, hazardous

production material handlers, and office areas utilizing computer workstations (e.g., CAD IC design or intensive data input/word processing). A brief discussion is presented on each of these areas, and the common risk factors for repetitive motion/material handling injuries.^[12]

4.1 Fab Area

The semiconductor fabrication area presents the potential for many of the risk factors previously discussed. The recently concluded semiconductor industry worker health study by the University of California at Davis investigated many of these factors closely in the various components of the study.^[13] They defined the primary ergonomic stressors of concern in wafer manufacturing as repetitive motion tasks, lifting activities, and long periods of standing. Specifically, they defined the following ergonomic stressors in relationship to their potential for causing adverse reproductive outcomes:

- Standing: based on the number of hours standing.
- Lifting: based on the number of times a subject lifted fifteen or more pounds a day.
- Constrained posture: based on the time spent looking through aligners, microscopes, or other inspection equipment with eyepieces without a break.
- Stress: based on a few simple questions, addressing how stressed one was at the end of the work day; how tired one was at pieces of equipment operated at one time.
- Shift: day, evening, or night.

In two of the four components of the U. C. Davis study, over half of the workers stated they spent six hours or greater standing each day, with less than 10% of the fab workers spending one hour or less per day standing. Standing is considered to be a static posture, and a risk factor both ergonomically and from a reproductive outcome perspective. Constrained postures such as might occur when using microscopes or other equipment with eyepieces was a common ergonomic exposure in the fab.^[13]

Approximately 30% of the workers studied identified that they normally spent two or more hours in a constrained posture without taking a break. Additionally, fab workers were more likely to report they were very cold or very hot in comparison to non-fab workers. When fab and non-fab workers were asked about the number of times per day they lifted items fifteen pounds or greater, there were no significant differences in the two groups.^[13]

Another task that is performed in the fab involves the “loading” of wafer boats that contain approximately 25 wafers. The individual wafers are hand transferred using a vacuum wand that involved holding down a thumb switch to activate the vacuum, and then a very precise transfer of the individual wafers. This job is also considered quite stressful, as the individual wafers themselves are highly breakable and, if the ICs on the wafer are far along in the process, highly valued (in the thousands of dollars in the case of microprocessors). Mechanized wafer transfer systems have helped to reduce the number of fab operators required to perform this repetitive manual transfer task.

Once the boats are loaded, they must either be transported within the sub-processing areas of the fab, or attached to the infeed of the next piece of processing equipment. In the case of diffusion furnaces, this may involve loading the wafers into an “elephant” which is made of thick quartz glass and involve substantial weights that may be required to be raised above the worker's shoulder height. This again presents risk factors for range of motion, force and posture. As larger diameter wafers are used, a fully loaded boat of 8 inch wafers can weigh in excess of ten pounds, and present a significant above-shoulder-height lifting hazard.

The U.C. Davis study^[12] looked closely at the musculoskeletal outcomes in the historical component of the study. They found that musculoskeletal outcomes were common, with prevalence in the nonfab group ranging from 2% (carpal tunnel syndrome) to 35 % (for back pain within the past year). Overall, 7 of 12 musculoskeletal outcomes were significantly more prevalent in fabrication room workers, with the adjusted odds ratios for these outcomes clustering around 1.5.

They also found that a composite musculoskeletal outcome “any distal upper extremity symptom” was increased among fab workers. A dose-response relationship was found with hours per day spent in the fab. The highest exposure group (six or more hours per day in fab) had a statistically significant odds ratio of 1.47 compared to the lowest category (less than four hours per day).

Work group analysis showed elevation of risk in several fab groups: furnaces (diffusion), etch, and maintenance. Risk was also elevated in persons working in both the etch and photo work groups. The odds ratio elevations on these work groups were comparable and ranged from 1.61 to 1.95. The agent-specific analysis indicated an association with increased lifting frequency, vacuum wand use, and use of the projection-aligner in photolithography. They note that the female gender was an important

co-variate of musculoskeletal symptom reporting, with odds ratios for female gender clustering around 2–3.

From a lifting perspective, the primary at-risk workers are the fab equipment or line maintenance personnel. They routinely are required to work in very tight spaces, and lift extremely heavy sub-components of process tools/pump packages of ion implanters, CVD, and EPI systems, metallization units (e-beam or sputterers), and etchers.^[14]

Another area of concern is the lifting (and in some cases the pouring) of heavy containers of process chemicals (solid metal targets, liquids, or gas cylinders primarily) that must be installed in secondary containment enclosures within very tight spacing. The use of bulk dispensing systems where chemical recharges are done by connection to large trailers or containers are recommended from an ergonomics perspective.^[15]

An area of growing concern within the newer generation fab areas relates to the presentation and positioning of the operator keyboards or control panels for process tools or control systems used within the fab. Many times, it appears the equipment manufacturer did not have ergonomics as one of the design features when the equipment was originally laid out. Also, the need for providing ergonomically designed chairs and workstations in the fab is an area of growing attention. The same design principles that are specified in this subsection for office environments are also applicable to the fab workstation.

4.2 Shipping and Receiving Docks

This exposure is not peculiar to semiconductor processing, but can be a substantial source of low back and neck injuries if appropriate manual material handling protocols are not observed, and mechanical equipment used.

4.3 Facilities Maintenance Shops

This exposure is also not peculiar to semiconductor processing, but maintenance technicians' job duties require the removal or movement of equipment, furnishings, structural materials, and supplies that are a substantial source of low back and neck injuries. The use of portable drills, portable screwdrivers, and other non-manual tools has helped to reduce the number of CTD's from these routine maintenance tasks.

4.4 Hazardous Production Materials (HPM) Handlers

Both the materials handlers in the chemical receiving/storage areas, and the hazardous waste handlers are presented with numerous risk factors in their routine job duties. They manually can be moving large drums of materials, some weighing as much as 500–800 pounds, and repetitively handling cases or bottles of HPM. Mechanical equipment in the form of lifts, pallet jacks, and drum lifters are needed to help reduce the number of lower back, neck, and upper extremity injuries.

4.5 Office Areas Utilizing Computer Work Stations

The largest relative number of semiconductor workers are exposed to the same hazard that many other information age office workers are experiencing, CTD symptoms, namely from intensive, repetitive interface with their computer workstation. A subset of this overall group of computer workstation operators are computer-aided design (CAD) engineers. These individuals typically spend eight hours plus per day working at an engineering workstation doing the layouts for the mask designs for the ICs. The reported incidence of upper extremity disorders has been on the increase since CAD engineers changed from the light pen as a input device, to the omnipresent mouse. In general, CAD users' risk factors are not appreciably different from the order entry person in sales, the accounts receivable clerk doing data entry, or the 80–120 words per minute administrative assistant doing word processing for an entire eight hour day. The same risk factors of posture, forceful exertions, and repetition are the ergonomic hazards that must be addressed, as well employee education and training, and mandatory work/stretching breaks.

5.0 ERGONOMIC DESIGN AND LAYOUT PRINCIPLES

Many of the basic principles of ergonomic design and layout are based on a common-sense approach to designing work tasks. The worker performing a task must be able to work in a comfortable position that allows a neutral posture, with a minimum of awkward postures and required movements. Additionally, to minimize the workers risk factors, minimize the number of repetitions per time period. Also, reduce the speed and force needed by the worker to perform the task. Finally, the best designs are

captive to the physical condition of the workers performing the task, so some type of conditioning or exercises are recommended to strengthen the hand, arm, back, neck, and shoulder muscles. This regular exercise may help to prevent injuries by reducing the need to compensate for weak or unconditioned muscles.

5.1 General Design and Layout Principles

Figure 7.5 provides recommendations for work place design and layout based on sound ergonomic principles.^[16] This is intended as a general “prompting”-type format for use in designing and laying out a work area or task.

5.2 Ergonomic Checklist for Office Work Stations

The following guidelines are provided on how to properly set up and use a computer work station. This ergonomic checklist offers guidelines for adjusting a chair and positioning the keyboard and display/monitor. It also includes suggestions for healthy work patterns, such as taking frequent, short, rest breaks. Of course, it is impossible to tell what’s comfortable for each individual, so each person has the ultimate responsibility for adapting and implementing these guidelines to meet individual needs.^[17]

CHAIR

- ___ Adjust chair height so feet are flat on the floor and with thighs parallel to the floor.
- ___ Chair back rest should support lower back and fit curvature of spine.
- ___ Chairs are available in different sizes for taller or shorter persons. One size does not fit all.
- ___ Allow clearance between front edge of seat pan and back of knees.
- ___ Adjust seat back tension for the persons weight in order to give proper back support.
- ___ Some chairs have other adjustments. The individual must learn how to use these features to provide optimal seating comfort.
- ___ Chairs should have a five-point (caster) support.

Ergonomic Principles Workplace Design and Layout	
Head height consideration: • Allow for the tallest • Natural posture is to look down slightly • Avoid narrow viewing angles	Knuckle height consideration: • Keep lifting tasks between knuckle and shoulder height
Shoulder height consideration: • Place items and parts between shoulder and waist height • Avoid reaches above the shoulder	Leg length consideration: • Allow for long legs • Provide adjustments or footrests for shorter legs
Elbow height consideration: • Normal work—raise surface to just below elbow height • Very precise work—raise the surface above elbow height and provide rests for arms • Heavy work—place work surface about 8 inches below elbow	Hand size consideration: • Allow for small hands for grasping • Allow for small hands for dangerous openings • Allow for large hands for access
Arm reach consideration: • Allow for the shortest when reaching up or out • Allow for the tallest when reaching down • Keep frequent work within forearm distance	Body bulk consideration: • Allow for the largest • Remember that clothing adds bulk
	Other items to check for: • Horizontal reach • Vertical reach • Leg room, twisting, upper body posture • Proper workplace design and layout reduces unnecessary stress
Factors in Design Guidelines for Adjustability	
A good fit between the person and the task can be obtained by making the workplace adjustable: • Raise or lower work surface • Change the worker's position • Use a tool to extend a reach • Adjust shape • Adjust location • Adjust orientation • Adjust chairs • Use support stools, swing bracket stools, and other props • Use platforms, step ups, and mechanical lifts	• Use footrests • Use armrests • Adjust design and location of tools • Adjust trays and carriers
	Adjusting the product or product containers: • Relocate or reorient product or work object • Use jigs, clamps, and vises • Use parts bins • Use lift tables, elevators, and similar equipment
Part Manipulation Considerations	
Range of movement: • Keep well within full range of movement • Never hyperextend any joint • Keep elbow at right angles when applying force • Keep wrists straight	Length of contraction: • Keep elbows to the side • Keep static muscular contractions (tensed muscles) to a minimum • Minimize time of muscular contraction • Minimize force of muscular contraction
Rate of repetition: • Allow frequent job rotation • Allow self-pacing (new employees should start off at a slower rate) • Keep rates to a minimum • Allow frequent rest pauses • Check fatigue in specific muscles	Force of application: • Never exceed 20% of operator's maximum force in a prolonged or repetitive work posture; avoid frequent impulses • Remember that legs, knees, hips, and shoulders may all be subjected to similar stresses

Figure 7.5. Recommendations for workplace design and layout based on sound ergonomic principles.^[16] (Reprinted with permission from the National Safety Council. National Safety Council (NSC). Ergonomics: A Practical Guide, 2nd ed. Itasca, IL: NSC, 1993)

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KEYBOARD, MOUSE, AND OTHER INPUT DEVICES

- ___ Adjust keyboard height with arms relaxed at sides, wrists straight, and forearms parallel to floor. Keyboard will be about elbow height.
- ___ If work surface height needs to be raised or lowered, request assistance from the appropriate workgroup.
- ___ Maintain clearance between bottom of work surface and top of thighs. Move the pencil tray to the side if necessary. Relocate file cabinets in order to lower as many sections of work surface as possible.
- ___ Position mouse and other input devices close by and at the same height as the keyboard.

VIDEO DISPLAY TERMINAL

- ___ Top of screen should be at or slightly below eye level. For large VDTs, try to keep objects low on the screen. Tilt/swivel VDT stands are available.
- ___ Avoid glare on screen from windows and overhead lights. Use blinds or curtains to block direct glare from windows. Turn screen at a 90° angle from window or light source. Tilt screen down to reduce bright spots from overhead lights. Antiglare screens are available to help reduce glare.
- ___ Set monitor controls for maximum contrast with minimal brightness.
- ___ Clean screen regularly.

ENVIRONMENT

- ___ Document holders should be used by those entering data or text from hardcopy. Position document holder near the VDT and at the same distance from the eyes as the VDT.
- ___ Keep items used frequently (e.g., telephone, manuals, etc.) within easy reach. Avoid reaching for heavy objects over shoulder height.

- ___ Avoid pounding or striking too hard on the buttons on the key board and mouse. Use a light touch. Relax fingers and shoulders while typing or using other input devices.
- ___ Take short, frequent rest breaks throughout the day. Occasionally do other types of work.
- ___ Relax your eyes by looking at a distant object.
- ___ Move your body. Stand and stretch occasionally.
- ___ Telephone headsets are recommended for individuals performing for extended time periods on the phone.
- ___ Bifocal lens wearers may need eye glasses that are specifically designed for computer work. Measure the distance between the individual's eyes and the monitor (typically 18 to 28 inches) and report this information to an optometrist.
- ___ Take steps to manage the stress in your daily life.
- ___ Ergonomic support tools such as document holders, adjustable keyboard trays, tilt/swivel VDT stands, antiglare screens, and input devices are very important in the overall alleviation of office ergonomic issues.

5.3 Work Place Design

Good workplace or workstation design will help reduce worker fatigue, exertion, and musculoskeletal disorders that can result from awkward postures and static loads. Good ergonomic design can also increase the efficiency and productivity of an operation. The general ergonomic principles listed should be considered in the design of any workstation.^[18]

- Keep elbows down—shoulder abduction angle no larger than 30 degrees (the angle between the upper arm and the torso, i.e., when the arm is raised sideways away from the body).
- Keep hands down—below shoulder height.
- Avoid long reaches—not over sixteen inches.
- Provide elbow supports for long reaches. Padded supports are preferred.
- Avoid using the first three inches of a work surface.

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- Avoid tilting head forward greater than 15 degrees from torso. Use head support if a greater angle is required.
- Avoid tilting the trunk forward.
- Avoid sharp edges on work surface.
- Visual scanning should require eye movements only. Head movements will contribute to fatigue.
- Allow for change in postures—at the worker's discretion (e.g., sit/stand work station).
- Avoid muscle overloading. The maximum force that can be exerted drops rapidly with time. When designing workplaces, the maximum exertion values for muscles can be found in most ergonomic workplace design reference texts.
 - for occasional activities, less than 50% of the maximum voluntary exertion.
 - for repeated activities, less than 30% maximum voluntary exertion.
 - for continuous application, less than 15% maximum voluntary exertion.
- Avoid foot controls for standing workers. If necessary, electric or pneumatic foot switches are preferred. Design for operation by either foot. Avoid elevated or mechanical foot actuating devices, but if they must be used, provide a foot rest at the same height as the foot control.
- Avoid hard floors for standing workers—supply floor mats, and provide a foot rest of up to eight inches high.
- Provide suitable seating.
- Provide suitable work surface height (adjustable, if possible).

6.0 SEMICONDUCTOR EQUIPMENT

6.1 SEMI S2-93, Safety Guidelines for Semiconductor Manufacturing Equipment^[19]

This document was developed by the collaborative work of SEMI and the input from SEMATECH's overall ergonomics task force in 1992.

The SEMATECH group efforts delineated the overall key principles of a total human-equipment output effectiveness.^[15]

- Make the user's contribution to the equipment output as efficient as possible so that the basic equipment output is compromised by human failures.
- Make combined user-equipment involvement as safe as possible so that neither human nor equipment failures will compromise the user's health or damage the hardware. Inherent in this objective is the avoidance of injury to others and of damage to adjacent hardware.
- Minimize stress that the equipment imposes on the user as he/she operates, services, or maintains it. This includes such stresses as an undue energy demand, frustration in trying to deal with the equipment at any point in the human-equipment interaction and worry about whether one is using the equipment properly.
- Maximize the acceptability of the equipment, not only in terms of its attractiveness, but also in terms of giving user the feeling that the equipment will allow them to use it efficiently and keep it in good working order with a minimum of effort.

The SEMI S2-1993 standard provides general guidelines on how to apply the standard for the design, installation, operation, maintenance, and service of semiconductor equipment:^[19]

- Equipment designed to prevent injury and equipment damage; minimize procedural errors.
- Operator workload, information processing requirements and physical demands do not exceed user capabilities or compromise safety.

The section in the standard on ergonomic hazards addresses the prevention of ergonomic-related safety issues by implementing sound human-factors engineering principles in design, and structuring job requirements around human performance capabilities and limitations. Ergonomic hazards should be designed-out or otherwise reduced to the maximum extent practicable during the design stage. The ergonomic hazard can exist whenever system design/installation results in task demands that exceed capabilities of trained personnel. Some key examples are:

- Controls that are confusing to operate
- Displays that are difficult to read/understand
- Operator or service technicians required to lift heavy or bulky components
- Operators required to perform repetitive motion or tasks to successfully operate the equipment
- Operators required to stand or maintain static or awkward postures
- Equipment is provided with poor access or inadequate clearances that require extended reaches or maintaining ergonomically stressful postures
- Operator or maintenance personnel are required to routinely reach, bend, or stoop to perform normal operation of the equipment

6.2 SEMI Document 2092, Ergonomics/Human Factors Guidelines for Semiconductor Manufacturing Equipment (draft)^[14]

This draft document provides basic ergonomics performance requirements and design criteria for semiconductor manufacturing equipment. Conformance to the guidelines will promote compatibility in the integration of users and equipment within a semiconductor manufacturing environment. Pertinent major sections within the document address:

- General Requirements
- Workstation Design
- Design for Maintainability
- Lifting/Strength and Materials Handling
- Warnings and Alarms
- Controls and Displays
- User-Computer Interface

One of the areas mentioned above that provides useful ergonomic information on strength, force, and materials handling includes:

- The application of force and/or lifting that exceeds the strength capabilities of the user may result in musculoskeletal stress and injury.

- Provide with hazard labels for components over 35 pounds.
- NIOSH publication 81-122 and computer aided methods can be used to effectively evaluate potential lifting hazards. Various published tables of psychophysical data (e.g., Snook, 1991) should be consulted to assess lifting demands.
- The percentage of a user's maximum strength exerted in execution of a task should conform with the following guidelines (utilize female strength data at 45 years or older):
 - exertions greater than 60% of the maximum lift capacity are to be avoided.
 - occasional effort, up to 50% , is acceptable.
 - frequent effort should be kept below 30%.
 - continuous effort should be kept below 15%.
- Handles or cutouts should be designed to facilitate use of the power grip (e.g., similar to that used to hold a pistol). Avoid handles that require pinch grips or deviated wrist postures.
- Lifting and handling tasks performed in a stooping position should be avoided.
- Two handed pushing or pulling tasks should be performed at or slightly below the elbow height.

7.0 SUMMARY

This chapter provides a brief overview of ergonomic principles and highlights some important topics such as carpal tunnel syndrome and office workstation design. Also, a review of the primary risks of semiconductor processing has been provided. Finally, general overall methods of ergonomics are based on a logical and systematic process of:

1. Establishing the proper role of the human with the equipment and work area
2. Designing the human-equipment interface to fit the human's capabilities and limitations
3. Evaluating and testing to see that the design does fit those capabilities and limitations

4. Properly training the human to operate the equipment or workstation, and understand the importance of the principles of ergonomics on their own individual body systems.^[3]

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